Mechanics of Human Movement

Course Progress **90.2%** (11 of 13)

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Week 9: Assignment 9

Due date on 2026-09-23, 23:59 IST

The due date for submitting this assignment has passed. Your last recorded submission was on 2026-09-18, 23:06 IST.

1. Which of the following is true for running? 1 Point

○ While running the GRF traces the butterfly curve similar to that in walking
● Running does not involve a double support phase
○ Stance phase is longer than swing phase for running
○ Gait is always asymmetric during running

Yes, the answer is correct. Score: 1
Accepted answer is Running does not involve a double support phase

2. Step width is a measure of the ________ separation of the foot 1 Point

● Mediolateral
○ Proximal-distal
○ Anterior-posterior
○ longitudinal

Yes, the answer is correct. Score: 1
Accepted answer is Mediolateral

3. The internal rotation of the thigh can be observed best in the 1 Point

○ Coronal Plane
○ Sagittal plane
● Transverse plane
○ Frontal plane

Yes, the answer is correct. Score: 1
Accepted answer is Transverse plane

4. Walking speed = ________ 1 Point

● Cadence * Stride length /120
○ 120 * Stride length /Cycle time
○ 120 * Cycle time /Cadence
○ Cadence * Stride length /Cycle time

Yes, the answer is correct. Score: 1
Accepted answer is Cadence * Stride length /120

5. Stance phase is approximately ______ of normal gait cycle. 1 Point

○ 40%
● 60%
○ 50%
○ 30%

Yes, the answer is correct. Score: 1
Accepted answer is 60%

6. What is the stride length of a person walking at the speed of 1 m/s and cadence of 120 steps/min? 1 Point

○ 0.5 m
○ 1.2 m
○ 1.5 m
● 1 m

Yes, the answer is correct. Score: 1
Accepted answer is 1 m

7. The maximum knee flexion is observed during _______ phase of walking. 1 Point

● Initial swing
○ Terminal stance
○ Terminal swing
○ Pre-swing

Yes, the answer is correct. Score: 1
Accepted answer is Initial swing

8. A gait cycle is made up of ________ 1 Point

☐ One step length
☑ Two step lengths
☑ One stride length
☐ Two stride lengths

Yes, the answer is correct. Score: 1
Accepted answer is Two step lengths, One stride length

9. Which phenomenon explains the M-shape of the GRF in the butterfly diagram? 1 Point

○ The movement of the COM in the transverse plane
● The upward and downward acceleration of the COM
○ The forward and backward acceleration of the COM along the direction of progression
○ The action of gravity

Yes, the answer is correct. Score: 1
Accepted answer is The upward and downward acceleration of the COM

10. What is the stride time of a person walking with step length of 0.6m and speed of 0.8 m/s? 1 Point

○ 0.5 s
○ 0.67 s
● 1.5 s
○ 1.2 s

Yes, the answer is correct. Score: 1
Accepted answer is 1.5 s

11. Which of the following events happen in single limb support? 1 Point

☑ Mid stance
☑ Terminal stance
☐ Initial contact
☐ Loading response

Yes, the answer is correct. Score: 1
Accepted answer is Mid stance, Terminal stance

12. Pre-swing phase of one limb corresponds to 1 Point

○ Terminal stance of other limb
○ Initial contact of the other limb
○ Mid stance of the other limb
● Loading response of the other limb

Yes, the answer is correct. Score: 1
Accepted answer is Loading response of the other limb

13. Which of the following phases involve the ankle rocker? 1 Point

○ Terminal Stance
● Mid stance
○ Loading response
○ Pre-swing

Yes, the answer is correct. Score: 1
Accepted answer is Mid stance

14. The maximum ankle plantarflexion happens during __________ phase of normal walking. 1 Point

○ Mid swing
○ Terminal swing
● Initial swing
○ Mid stance

Yes, the answer is correct. Score: 1
Accepted answer is Initial swing

15. The distance between successive contacts of the same limb with the ground is called as__________. 1 Point

○ Step length
● Stride length
○ Cadence
○ Double support

Yes, the answer is correct. Score: 1
Accepted answer is Stride length